

TRANSCRIBE PROJECT DOCUMENTS

Technical Requirements Document (TRD)

Document 02

Status	Draft v1
Related	01 PRD , 04 App Flow , 05 Backend Schema

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1 Architecture overview

Principles:

1. **Human text is authoritative.** Model output is stored as a *hypothesis* next to reviewer text and is never written over it.
2. **Immutability.** Source audio and transcripts are content-addressed (sha256) and never modified. Datasets are frozen snapshots, and model versions point to exactly one snapshot.
3. **One process model.** A single user on CPU needs no Redis or Celery. A worker process polls a jobs table in SQLite.
4. **Engines behind interfaces.** Aligner, VAD and ASR are swappable: a Protocol plus a registry, so the CTC comparison model or a larger Whisper plugs in without UI changes.

2 Model strategy

2.1 Why not “a proper English model + a proper Swahili model, blended”

This was the proposal in the requirements discussion. It does not work for Kenyan code-switching, for three reasons:

Local machine (CPU)

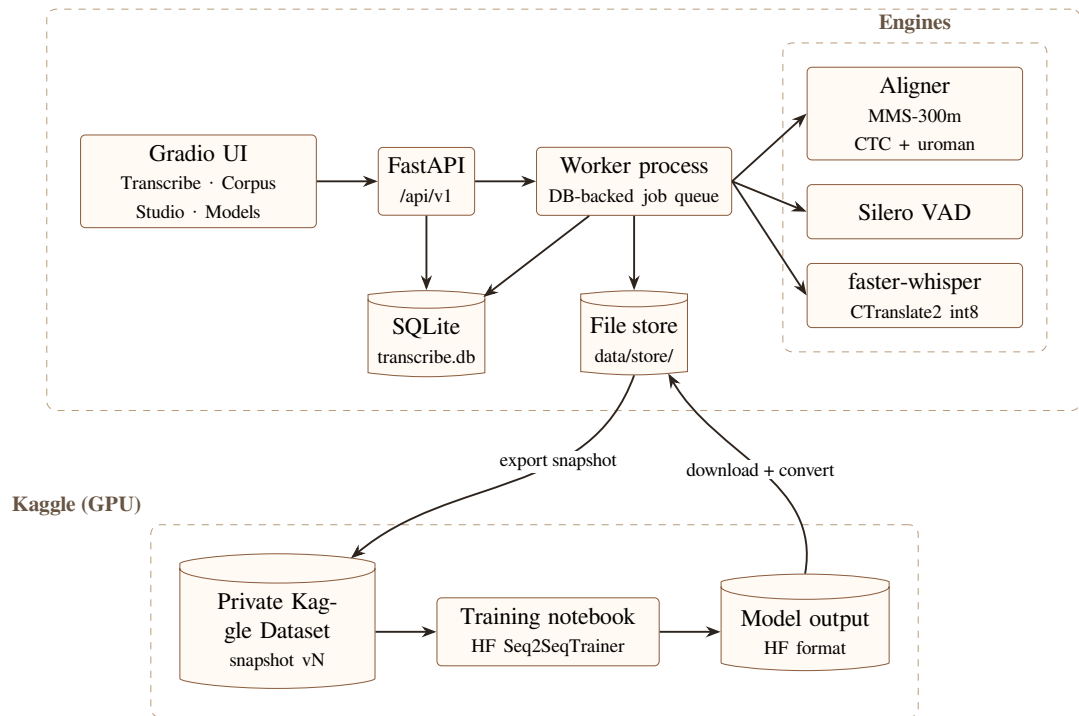


Figure 1: System architecture: local CPU services and Kaggle GPU training.

- **Routing needs per-word language ID before transcription.** A router (language ID → English or Swahili model) decides per segment. Switches happen mid-sentence and *mid-word* (*nimedownload*), below any practical language-ID granularity. Every routing error becomes a transcription error.
- **Monolingual models cannot output the other language.** A Swahili model spells English words in Swahili orthography (*daunlodi*), and an English model maps Swahili words onto English lookalikes. Merging hypotheses (ROVER-style) needs both hypotheses to contain the right word somewhere, and for intra-word switches neither does.
- **Cost and complexity.** It needs twice the compute on CPU, two fine-tuning tracks, and a merge heuristic with no training signal.

What we do instead: one multilingual pretrained model whose pretraining already covered both languages, fine-tuned on Kenyan code-switched speech. The blending happens in two places where it works:

- **The data mixture.** In-domain code-switched segments, plus public Kiswahili and public Kenyan-accented English, blended by sampling weights. This teaches the switch and keeps both monolingual abilities.
- **The tokenizer.** Whisper's byte-level BPE can emit any Latin string, so *nime + download* is just a token sequence. Nothing forces a single language per word.

2.2 Chosen models

Role	Model	Why
v1 production	openai/whisper-small (244M), full fine-tune	Fits a Kaggle T4 in fp16. Readable output with casing and punctuation. Its 30 s window matches our segments. Mature CPU serving via faster-whisper int8. The fast iteration suits learning the process

Role	Model	Why
v2 candidate	openai/whisper-large-v3-turbo (809M, 4-layer decoder), LoRA or 8-bit optimiser	Much stronger encoder with an inference cost close to small/medium. Adopted only if it beats v1 on the frozen dev set by a meaningful margin
Comparison (learning track)	facebook/w2v-bert-2.0 + CTC head, optional KenLM trained on code-switched text	Contrasting paradigm: no hallucination, faithful verbatim output, character-level. Teaches CTC fine-tuning and LM fusion. Not production because it has no punctuation and weaker CPU tooling
Aligner (not trained)	MMS-300m forced-alignment model + uroman romaniser (via ctc-forced-aligner, or torchaudio MMS_FA pinned)	Language-agnostic: it aligns any romanised text, so mixed EN/SW transcripts align without choosing a language
VAD	Silero VAD	Robust on noise and music compared with the current energy threshold. Runs on CPU in ms per window
Legacy	Existing MFCC CNN-LSTM CTC/Seq2Seq	Kept under legacy/ as an educational baseline only. Not deployable (see §2.4)

Uncertainty on the aligner: torchaudio has been deprecating parts of its API. The aligner sits behind an interface and its dependency is pinned, so either backend can be used.

2.3 Whisper fine-tuning specifics

- **Language token:** fixed `<|sw|>` + `<|transcribe|>` + `<|notimestamps|>` for all training samples. Kenyan code-switching is predominantly Kiswahili-framed, and a fixed token removes the language-ID step at inference. **Experiment E3** compares sw vs en vs no language token on the dev set. This is a cheap and instructive ablation.
- **Non-speech examples:** about 5% of training samples are music/noise/silence windows with an empty target. This is the most direct fix for church and TV hallucination.
- **Data mixture (v1):** in-domain approved segments weight 1.0. Public Kiswahili (Common Voice sw, FLEURS sw_ke) weight about 0.3, to prevent forgetting and add speaker diversity. Kenyan-accented English (AfriSpeech-200 Kenyan subset) weight about 0.2. Licences must be checked before any commercial use (AfriSpeech is non-commercial).
- **Hyperparameters (starting point, T4 16 GB):** fp16, lr 1e-5, warmup 500, per-device batch 16 with gradient checkpointing, effective batch 32, 3–5k steps, evaluate every 500 steps on dev WER, keep the best checkpoint. SpecAugment is on (mask_time_prob=0.05).
- **Kaggle constraints:** a weekly GPU quota (~30 h) and a hard per-session limit, so we save checkpoints to /kaggle/working every 500 steps, the training script supports `resume_from_checkpoint`, and one session does one experiment.

2.4 Why the legacy model cannot be the product

- Hop 512 at 16 kHz gives 32 ms frames, and the three pooling layers make that 256 ms per CTC output frame. A 30 s segment has about 117 frames against about 400 target characters, and CTC requires frames $\geq$ target length. `CTCLoss(zero_infinity=True)` silently zeroes these losses.
- The tokenizer removes spaces, so output cannot be readable.
- It has no pretraining, and hundreds of hours of audio are needed before from-scratch ASR becomes competitive with a fine-tuned pretrained model.

If kept for learning, fix it with hop 160 (10 ms), 2× downsampling, a space token, and no SOS/EOS in CTC targets.

3 Segment length policy

(Answers “what chunk length?”)

Parameter	Value	Reason
Max segment	30.0 s	Whisper's encoder window is exactly 30 s; anything longer is truncated
Target	8–20 s	Enough context for the decoder; still fits CTC memory for the comparison model
Min segment	1.5 s	Shorter segments carry little context and add label noise
Split points	Aligned inter-word gaps ≥ 300 ms; prefer the largest gap; prefer turn boundaries	Never cut a word; natural prosodic units
Padding	100 ms on each side, clipped to the neighbouring word midpoint	Avoids clipped onsets without leaking neighbouring words
Speaker	A segment never spans a speaker turn change unless the turn is < 1.5 s	Clean targets; overlap is flagged

At inference, Silero VAD speech regions are merged into windows of ≤ 30 s. Windows are cut at the longest silence and overlap by 0 s (VAD-cut, so no stitching is needed).

4 Corpus pipeline

ingest → normalise audio → parse transcript → normalise text for alignment→
 long-form alignment → word timestamps + scores → segmentation→
 auto-flags + priority score → review → approve → dataset snapshot

4.1 Ingest

- ffmpeg decodes to a 16 kHz mono 16-bit WAV. The original file is kept byte-identical, stored at data/store/originals/<sha256>.<ext>.
- The transcript is parsed by the existing docx_parser / pdf_parser. The speaker regex and language hints (italic → sw, a convention that is verified per source) are reused.

4.2 Text normalisation for alignment (a separate text, never shown as the label)

- Remove bracketed annotations ([unclear], [laughs], [crosstalk]), but record their word positions for flags.
- Lowercase, and romanise with uroman (a no-op for Latin script, but it normalises diacritics and apostrophes).
- Numbers are verbalised ([STYLE_GUIDE §4](#)). For each digit token the pipeline generates an English and a Kiswahili candidate (the default comes from the span language, years → English, labels → English title case). It aligns both candidates locally and keeps the one with the higher score. If the scores differ by less than a margin (initially 0.1), the word becomes the * wildcard and the segment is flagged number_check. The chosen spoken form becomes the segment label. English uses num2words; Kiswahili uses a small rule-based verbaliser in corpus/numbers_sw.py (the system is regular, and it gets unit tests from the style guide table).
- Optional display ITN (words → digits) runs on Transcribe output only (P1), never on training text.
- Keep a word-level index map norm_word_i → source_char_span so aligned timestamps map back onto the **original** transcript text, which becomes the segment label.

4.3 Long-form alignment

- CTC emissions are computed over the full audio in 30 s windows with 2 s overlap, then stitched.
- Alignment is turn-anchored: each transcript turn is aligned inside a search window around its expected position (moving anchor). This keeps the Viterbi trellis small ($T \times L$ per turn instead of per recording) and localises damage from non-verbatim passages.
- Output per word: `start_s`, `end_s`, and `score` (mean frame log-posterior, turned into a 0–1 confidence).
- A turn whose mean confidence is below θ (initially 0.5, tuned on the first reviewed recording) is flagged `low_alignment`. That is the usual signature of a non-verbatim or missing passage.

4.4 Priority score for the review queue

```
priority = 0.5 · (1 - alignment_conf) + 0.2 · has_unclear_marker + 0.1 · overlap_marker
          + 0.2 · model_disagreement (WER(aligned_text, model_hypothesis), once a model exists)
```

The review queue sorts by priority descending. Segments with high confidence and low disagreement can be **batch-approved after spot-checking** a random 10%, controlled by a setting and off by default.

4.5 Recordings without transcripts

Silero VAD splits them into ≤ 30 s chunks, the current production model pre-fills the text (stored as a hypothesis, with the reviewer text starting from a copy), and they go into the same review UI.

5 Inference pipeline

1. Decode with `ffmpeg` → 16 kHz mono float32.
2. Silero VAD (threshold 0.5, min speech 250 ms, min silence 400 ms) gives speech regions.
3. Merge regions into windows of ≤ 30 s.
4. Run faster-whisper on each window: `language="sw", task="transcribe", beam_size=5, condition_on_previous_text=False, word_timestamps=True, vad_filter=False` (VAD has already been applied).
5. Guards: drop segments with `no_speech_prob > 0.6` and `avg_logprob < -1.0`. Drop segments with compression ratio > 2.4 (repetition loops). Collapse a 3-gram repeated more than 3 times.
6. Post-processing: timestamps are offset back to absolute time, and segments are persisted as `job_segments`.
7. Export renderers: TXT, SRT, VTT, JSON, DOCX.

Batching: windows run sequentially, with `cpu_threads = physical cores` and `compute_type="int8"`. Progress = processed audio seconds / total speech seconds.

6 Service design

6.1 Processes

Process	Command	Responsibility
api	<code>python -m transcribe.api</code>	FastAPI + Gradio mounted at <code>/</code> . Handles CRUD and enqueues jobs
worker	<code>python -m transcribe.worker</code>	Polls jobs (<code>status='queued'</code>), runs alignment, transcription, evaluation and export, and heartbeats

Both processes share SQLite in WAL mode. A job is claimed with `UPDATE jobs SET status='running', worker_id=? WHERE id=? AND status='queued'`. A stale heartbeat (> 2 min) puts the job back in the queue.

6.2 REST API (/api/v1)

Method	Path	Purpose
POST	/recordings	Multipart upload of audio (+ transcript), domain, consent_scope
GET	/recordings, /recordings/{id}	List and detail, with review progress
POST	/recordings/{id}/align	Enqueue alignment + segmentation
GET	/review/queue?recording_id=&limit=	Next segments by priority
GET	/segments/{id}	Segment + current revision + hypotheses + audio URL
POST	/segments/{id}/revisions	Create a new revision (text, bounds, flags, spans, status). Rejected with 409 on version conflict
POST	/segments/{id}/hypothesis	Run the model on the segment (stored as a hypothesis only)
GET	/segments/{id}/audio	WAV slice (rendered on demand from the recording, cached)
POST	/datasets	Build a snapshot from a filter spec + split spec
POST	/datasets/{id}/export	Write the HF-format package for Kaggle
POST	/models	Register a model (upload or path), enqueue CT2 conversion
POST	/models/{id}/promote	Set the production model
POST	/evaluations	Enqueue evaluation of a model on a dataset split
GET	/evaluations/{id}	Metrics + per-segment results
POST	/transcriptions	Upload a file for transcription (enqueue a job)
GET	/transcriptions/{id}	Status, progress, segments
PATCH	/transcriptions/{id}/segments/{seg_id}	Save an edit
GET	/transcriptions/{id}/export?format=srt	Download
POST	/transcriptions/{id}/promote	Copy edited segments into the corpus (consent-checked)

Optimistic concurrency: every revision POST carries `base_revision_id`. A mismatch returns 409, so a second browser tab cannot overwrite silently.

6.3 Package layout (target)

```
src/transcribe/
  api/          # FastAPI app, routers, schemas (pydantic)
  worker/       # job loop, handlers
  db/           # SQLAlchemy models, session, Alembic migrations
  corpus/
    ingest.py   parsers/ (docx_parser.py, pdf_parser.py – moved from preprocessing)
    textnorm.py align.py segment.py priority.py
  inference/
    vad.py      engine.py (Protocol + faster-whisper impl) longform.py guards.py export/
  datasets/     # snapshot build, grouped split, HF export
  finetune/     # whisper_train.py, w2vbert_ctc_train.py (run inside Kaggle)
  evaluation/   # normalizer.py, metrics.py (WER/CER/switch-point/hallucination), report.py
  ui/           # Gradio Blocks: transcribe.py, review.py, models.py, theme.py
  legacy/       # ctc_model.py, seq2seq.py, trainer.py, data/ (unchanged behaviour)
notebooks/kaggle/ # thin notebooks calling transcribe.finetune.*
tests/
```

The package imports as `transcribe` (installed via `pip install -e .`). The `sys.path / src.` import hack is

removed.

7 Dependencies

Group	Packages
core	fastapi, uvicorn, sqlalchemy≥2, alembic, pydantic≥2, python-multipart, soundfile, numpy, ffmpeg (system)
corpus	python-docx, pypdf, etc-forced-aligner (or torchaudio pinned), uroman, torch (CPU)
inference	faster-whisper, silero-vad (onnx), python-docx (export)
ui	gradio≥4
eval	jiwer
finetune (Kaggle only)	transformers, datasets, accelerate, evaluate, peft (v2), bitsandbytes (v2)
legacy	librosa, scikit-learn, editdistance, nltk

Requirements are split into requirements/{core,corpus,inference,ui,finetune,legacy}.txt. Versions are pinned with upper bounds, following the current practice.

8 Evaluation

8.1 Splits (answers “how should we split?”)

- **Group by recording, never by chunk.** Chunks from one recording share speaker, microphone, room and topic, so a random chunk split leaks all of them and inflates results.
- If a speaker appears in several recordings (the same interviewer), group by speaker cluster: all recordings of that speaker go to the same split, or that speaker is held out explicitly.
- **Size dev/test in hours and recordings, not percentages.** Evaluation sets only need to be big enough for a stable WER, and variance comes from speakers and recordings more than from word count. Target: **test $\approx$ 3 h from $\geq$ 15 recordings, dev $\approx$ 2–3 h from $\geq$ 10 recordings.** Everything else is train. With 40 h that is about 85/7/8; as audio keeps arriving, train grows and dev/test stay fixed.
- **Test is frozen forever** (for comparability across model versions), stratified so it contains clean and noisy, low and high code-switch density, and each domain available. Dev is used for checkpoint selection and ablations.
- **Audio still arriving:** fix test_v1 and dev_v1 from the recordings available at the end of Phase 2, and never change them. New recordings go to train by default. A new domain or a clearly new condition (TV, church, a new region) first contributes $\approx$ 1 h to an external_<domain> test set.
- New recordings go to train by default. New *domains* (TV, church) first get a held-out evaluation set of $\geq$ 1 h before any of their data goes into train.
- Also keep FLEURS sw_ke test as a forgetting check. Test-set segments are never used for anything except final reports.

8.2 Normaliser (versioned, normalizer_version stored with each evaluation)

Lowercase. Strip punctuation except intra-word apostrophes. Remove hyphens (so *nime-download* $\equiv$ *nime-download*). Collapse whitespace. Remove bracketed annotations. Map a small table of spelling variants (e.g. *ok/okay*) from the style guide. Verbalise any remaining digits with the same rules as the corpus pipeline, so digit and word outputs score the same.

8.3 Metrics

Metric	Definition
WER / CER	jiwer on normalised text, micro-averaged (total errors / total reference words)
Per-language WER	Using reference language spans: errors attributed to en, sw and mixed reference words via the jiwer alignment
Switch-point error rate	Over reference words adjacent to a language change (plus all mixed words): the fraction misrecognised. This directly measures the code-switching capability
Hallucination rate	Words emitted per minute on reference non-speech segments (the non-speech test subset)
RTF	Wall time / audio duration on the reference CPU

Reports are broken down by domain and by flag (clean / noisy / overlap).

9 Security and privacy

- The app binds to 127.0.0.1 by default. Exposing it on a network requires setting TRANSCRIBE_AUTH_TOKEN (a bearer token middleware) in v1.
- `consent_scope` $\in$ {training, eval_only, transcription_only} on every recording is enforced in the dataset builder (training split only from training), in the Kaggle export, and in “promote to corpus”.
- Kaggle datasets are created private. The export package contains only audio slices and text, never names from the transcript header or file paths.
- No demographic or ethnicity fields are stored. Speaker labels are opaque (S1, interviewer).
- Uploaded transcription files are deleted after N days (configurable) unless promoted.

10 Testing strategy

Level	What
Unit	Text normaliser, segmenter on synthetic word timings, priority, grouped split (asserts no recording in two splits), metric functions (hand-computed cases incl. switch points), guards
Integration	A 60 s synthetic fixture (TTS or public-domain clip + transcript) goes through align → segment → dataset export, all on CPU in < 2 min
API	FastAPI TestClient: revision conflict (409), consent enforcement, job lifecycle
Model	The Kaggle notebook prints dev WER. <code>evaluate</code> reproduces the same WER locally within ± 0.5 absolute
CI	GitHub Actions: ruff + pytest (unit + integration with a tiny aligner mock) on Ubuntu, Python 3.10